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SESSION NAME

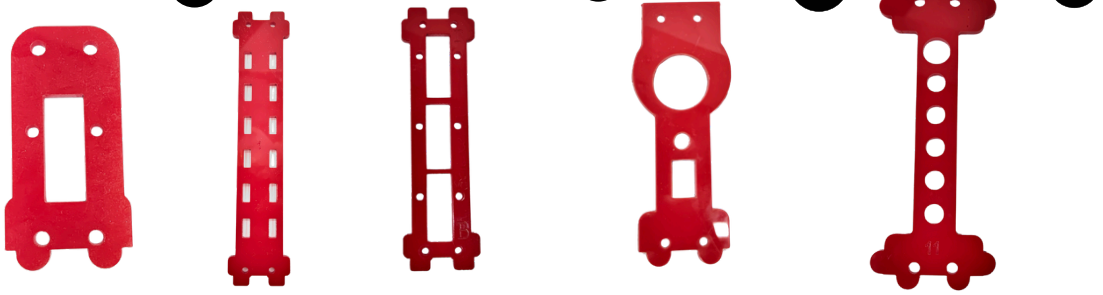
Session

1

STRUCTURE AND FASTENERS OF MICROCONTROLLERS

1 2 3 4 5 6

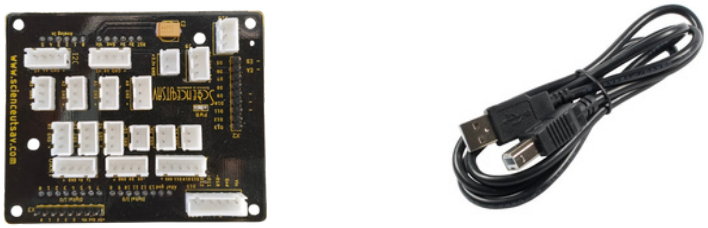
7 8 9 10 11 12



1 2 3 4 5

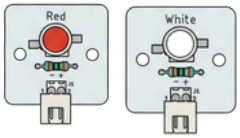
Spacers | L Clamp | Spacer Screws | Nuts | Screw

6 7



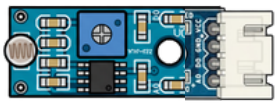
Arduino With Shield | Arduino cable

ELECTRONIC COMPONENTS

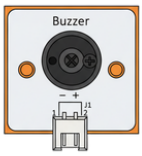


1 Watt LED PCB

1 Watt LED is a high-brightness light-emitting diode that produces much stronger light compared to normal small LEDs.



LDR sensor- A 4-pin LDR sensor is a light-sensing module that uses a Light Dependent Resistor (LDR) to detect brightness levels in the environment. It changes its resistance based on the amount of light falling on it—low resistance in bright light and high resistance in darkness. The 4-pin module typically includes extra circuitry (like a comparator) and has pins for VCC, GND, digital output (DO), and analog output (AO).



BUZZER PCB

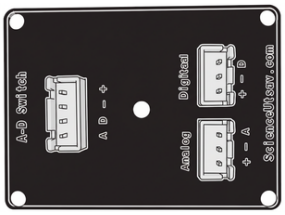
A buzzer is an electronic component that generates sound when electric current flows through it. It is commonly utilized in alarms, alerts, and notification systems to provide auditory signals.

STRING LED

A string LED is a series of multiple LEDs connected together in a chain to produce decorative or continuous lighting.
These LEDs are arranged in a sequence and powered by a single source, allowing them to glow together or in patterns.

Push Switch

A push switch is a button-style switch that operates only when pressed.
It provides temporary input in circuits, such as starting a device or triggering an action.



AD SWITCH PCB

The AD SWITCH PCB features two outputs: one analog and one digital. Users can easily select their desired output.

3 PIN RMC

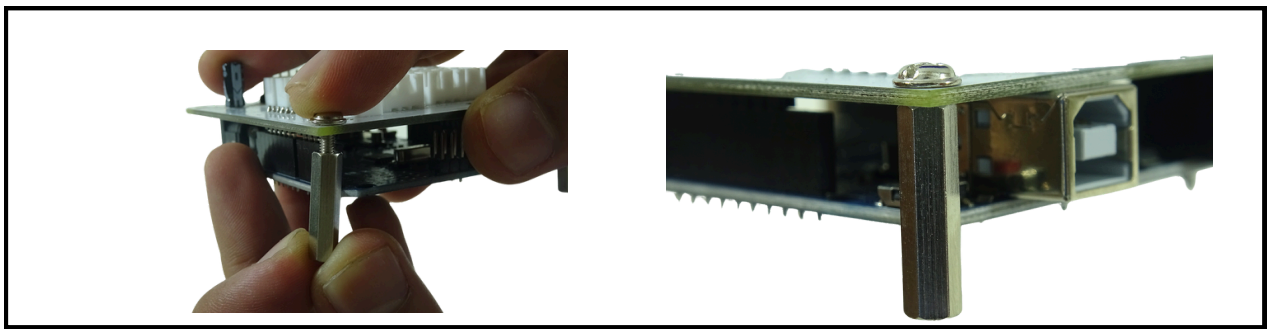
Used to connect the Sensor from AD switch to Shield

2 PIN RMC

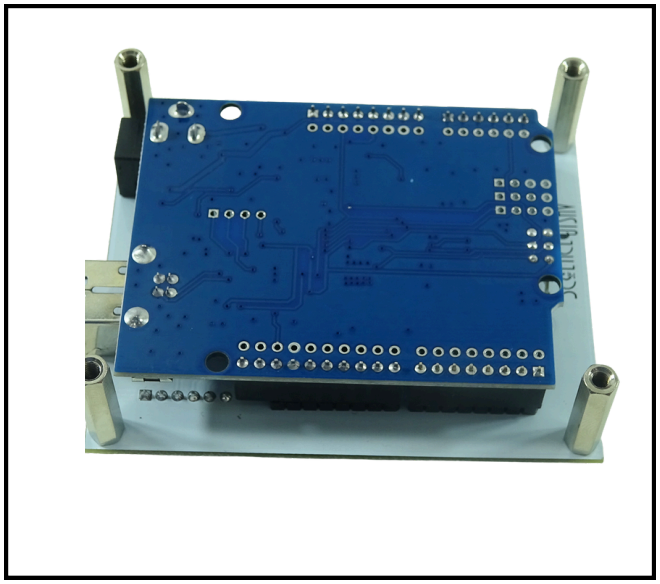
Used to connect the 1-watt LEDs and the buzzer.

Procedure :

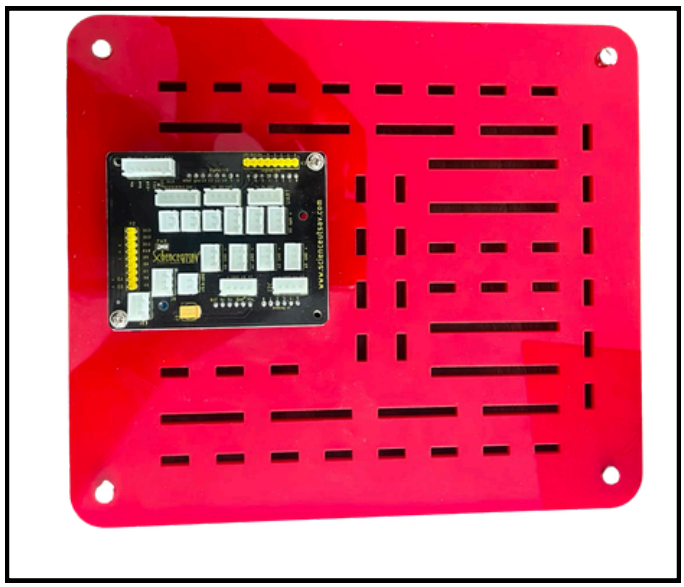
Step 1 : Fit the spacer to Arduino Shield as shown below using Spacer screw



Step 2 : Repeat the step 1 for all 4 corners of Arduino Shield



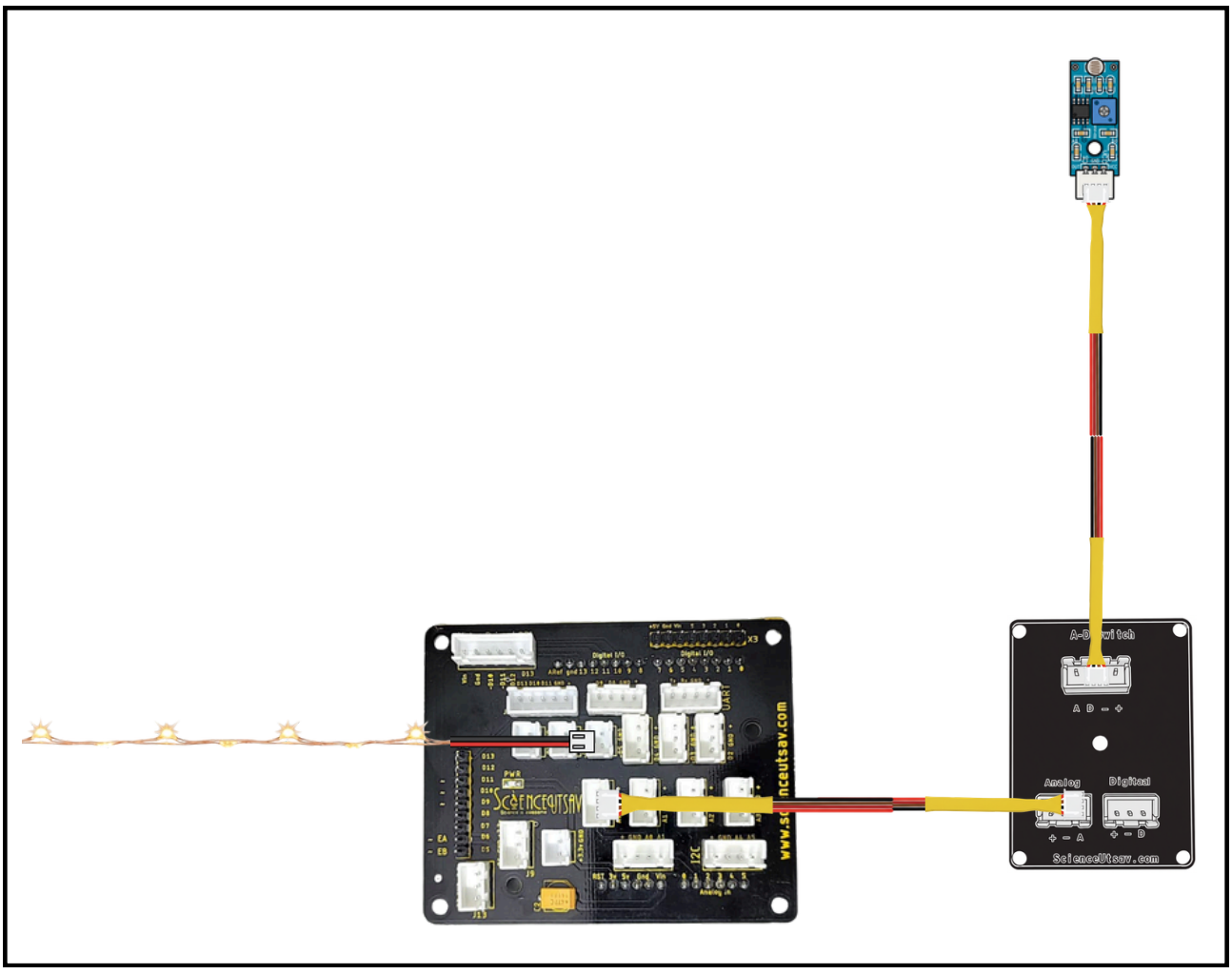
Step 3 : Place the Arduino Shield to the base board as shown below



Step 4 : Fix it with the spacer screw from the bottom as shown below

BEGINNER PROJECT

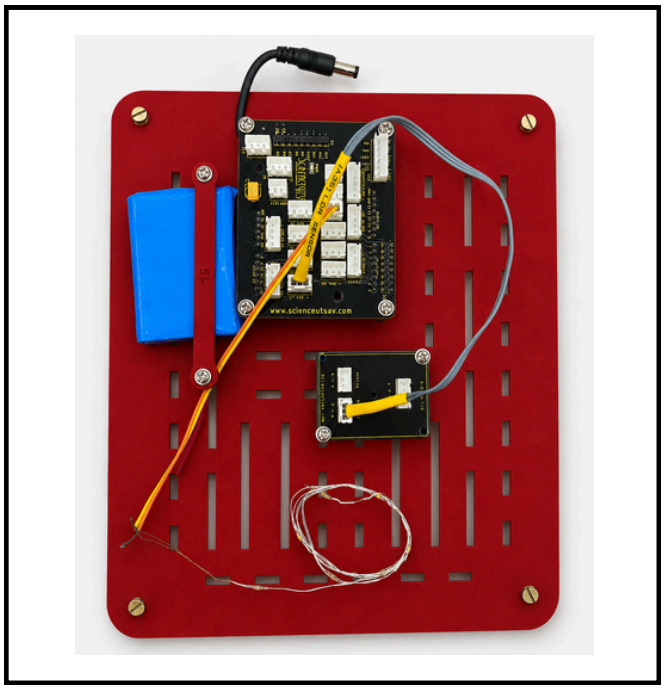
The circuit diagram of the project is given below



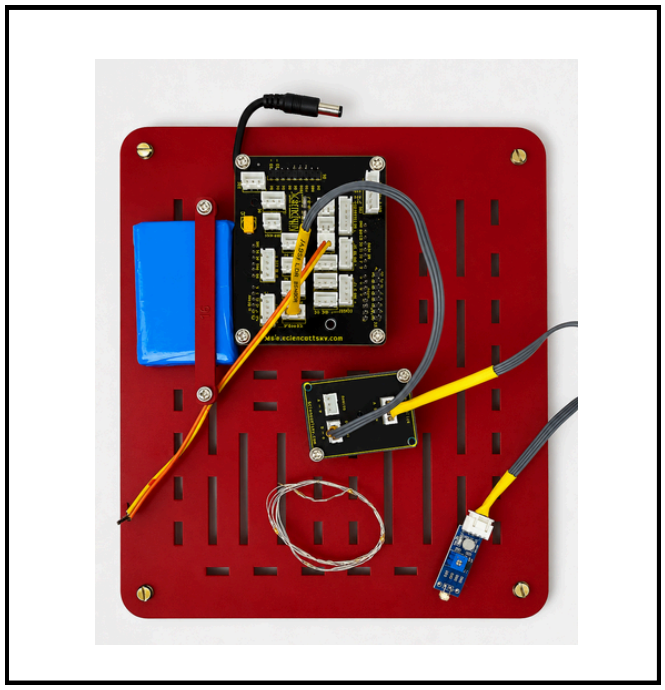
CONNECTION :

1. A0 - AD SWITCH - LDR SENSOR
2. D5 - STRING LED

Step 1 – Connect the string LED along with AD switch to the shield



Step 2- Connect the LDR sensor to the AD switch as shown



Step 3- Take L clamp and attach template in to it as shown in figure

Step 4- Insert the string LED to the template as shown

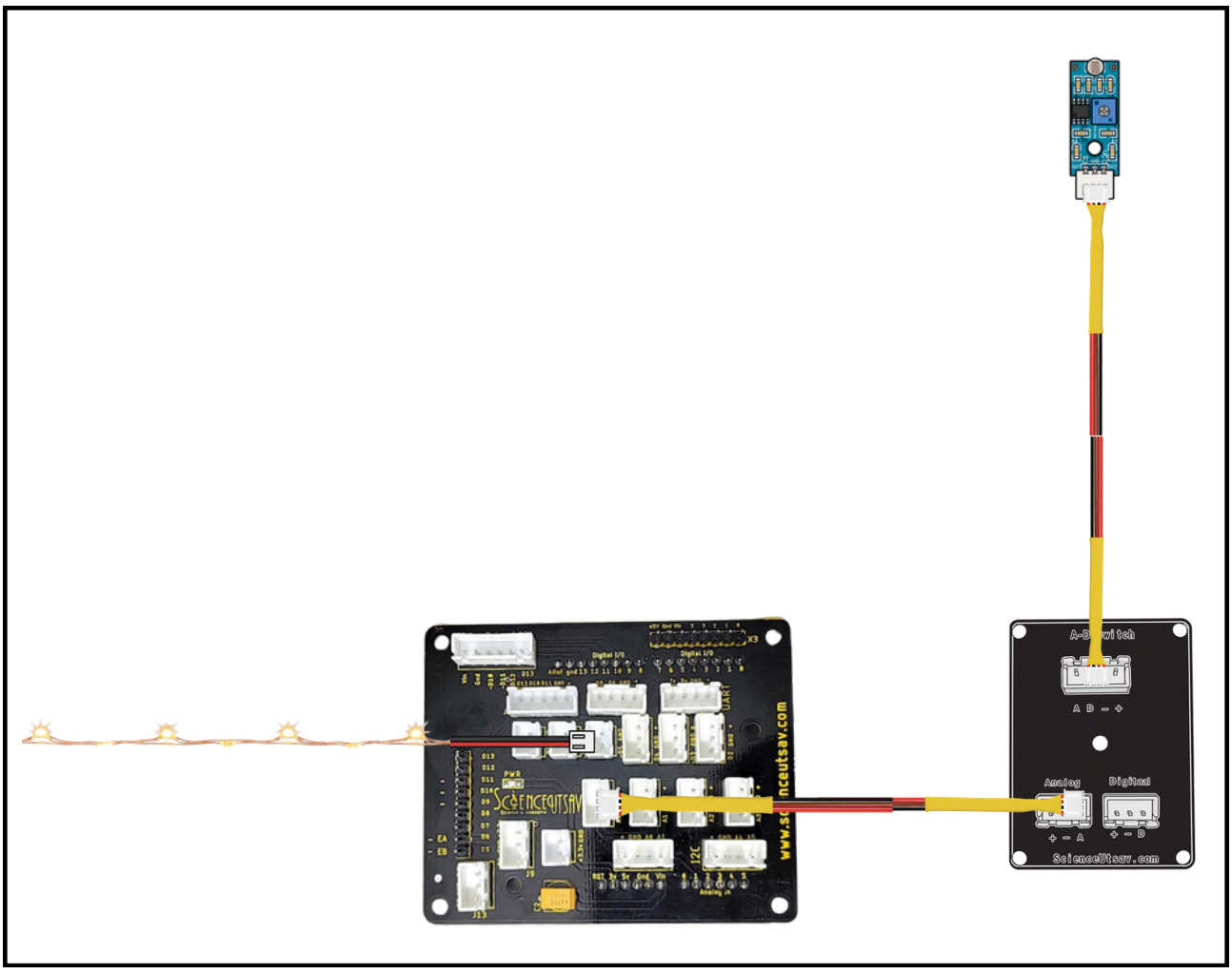
Step 5- Attach the template to the stand as shown

Code

```
// Session 1 Grade 6
// Sheep / Cow - LDR Auto String LED

const int ldrPin = A0;
const int stringLED = 5;

int darkValue = 500; // adjust after testing

void setup() {
 pinMode(stringLED, OUTPUT);
}

void loop() {
 int ldrValue = analogRead(ldrPin);

 if (ldrValue > darkValue) {
 digitalWrite(stringLED, HIGH); // dark = LED ON
 } else {
 digitalWrite(stringLED, LOW); // bright = LED OFF
 }
}
```

BEGINNER PROJECT

The circuit diagram of the project is given below



CONNECTION :

1.A0 - AD SWITCH - LDR SENSOR

2.D5 - STRING LED

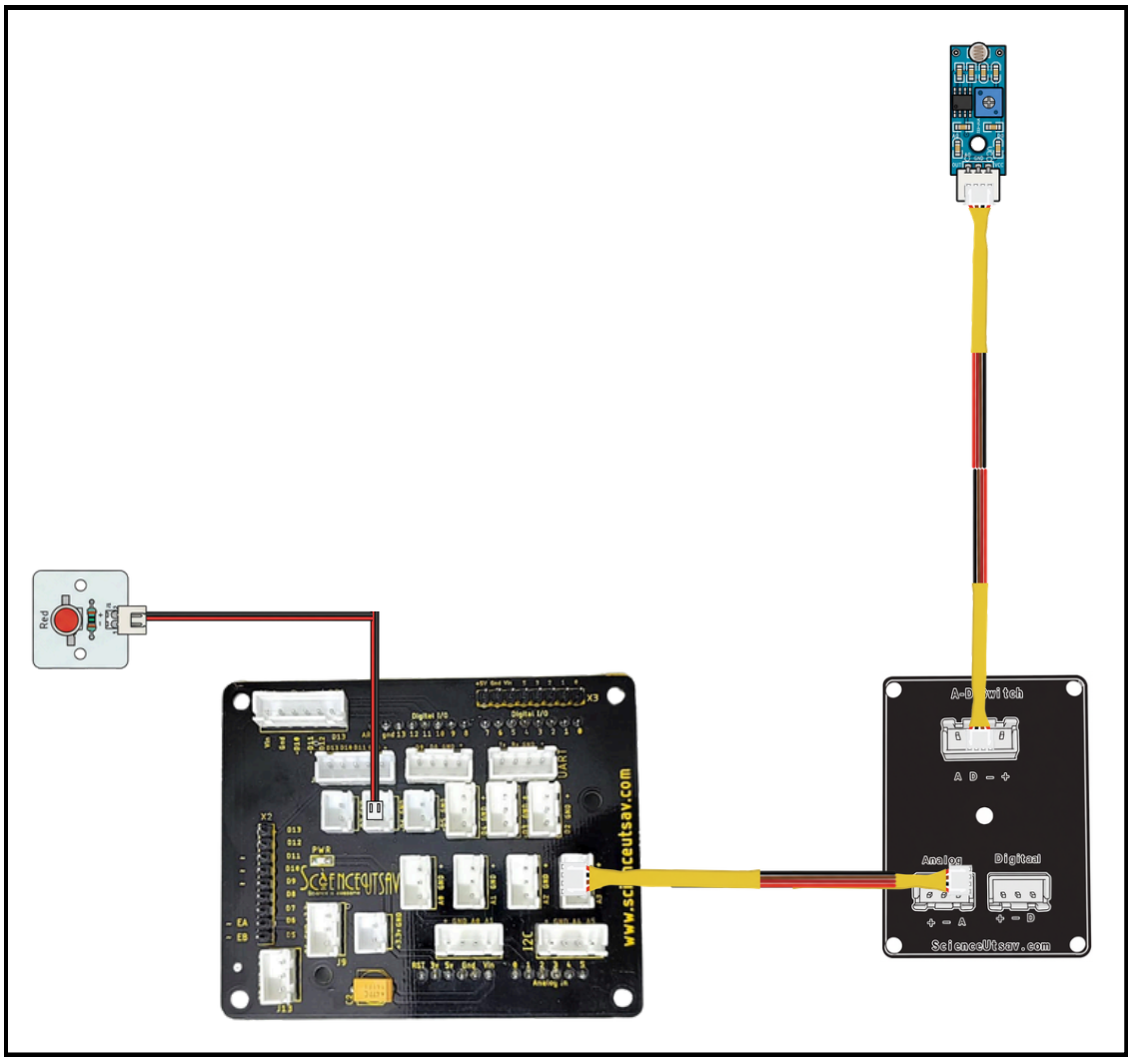
Follow the same procedure as before for this project

Code

```
// Session 1 Grade 6
// Sheep / Cow - LDR Auto String LED

const int ldrPin = A0;
const int stringLED = 5;

int darkValue = 500; // adjust after testing

void setup() {
 pinMode(stringLED, OUTPUT);
}

void loop() {
 int ldrValue = analogRead(ldrPin);

 if (ldrValue > darkValue) {
 digitalWrite(stringLED, HIGH); // dark = LED ON
 } else {
 digitalWrite(stringLED, LOW); // bright = LED OFF
 }
}
```

INTERMEDIATE PROJECT

The circuit diagram of the project is given below



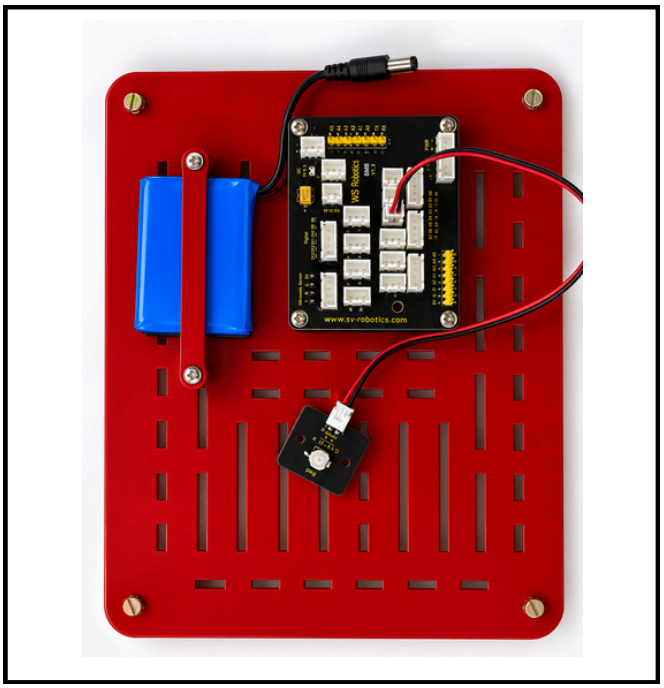
CONNECTION :

1.) A0 - AD SWITCH - LDR SENSOR

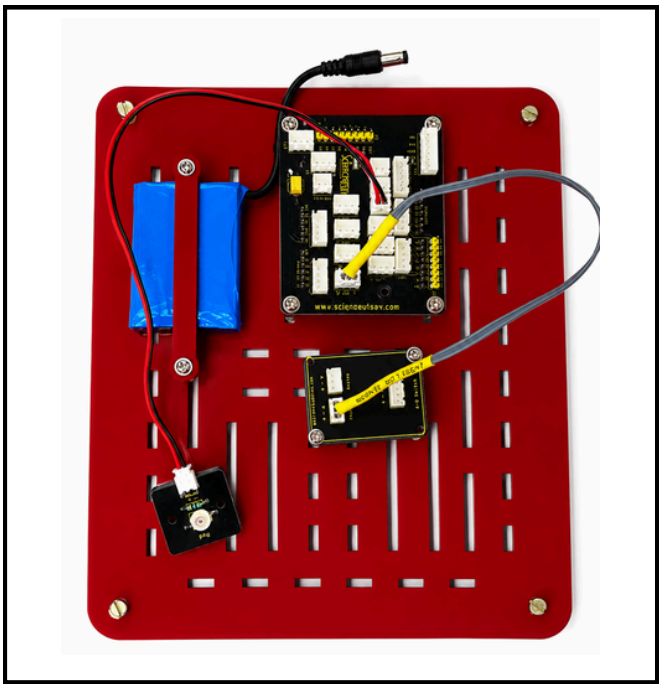
2.) D6 - 1W LED

Step 1- Attach 4 pin LDR sensor to the template with nylon screw and nut as shown

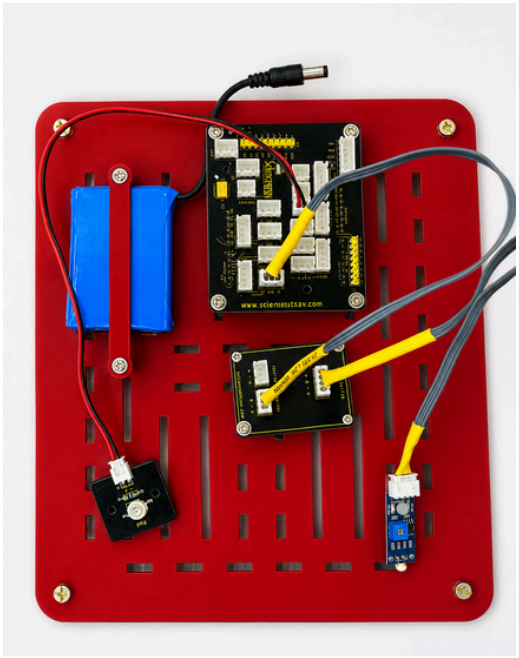
Step 2- Attach the 1W LED to the board



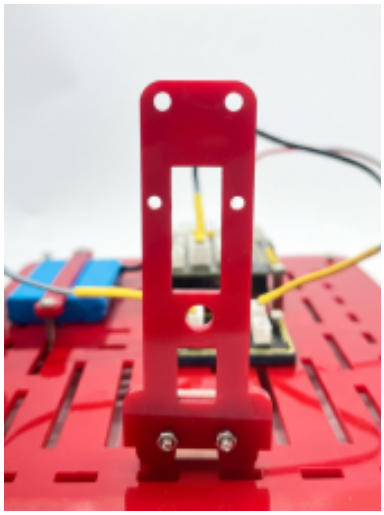
Step 3- Attach the AD switch as shown



Step 4- Connect the LDR sensor to the AD switch

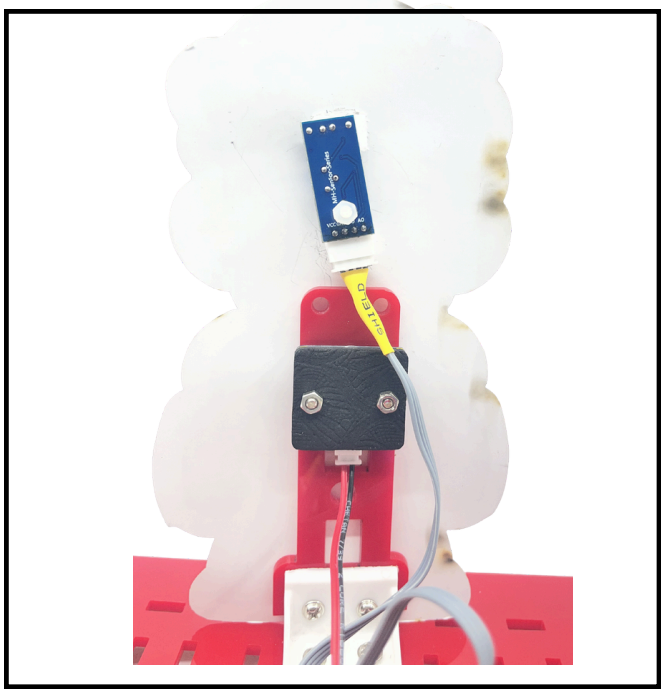


Step 5- Fix the stand to the board



Step 6- Attach the template to the stand

Step 7- Fix the LED to the template as shown



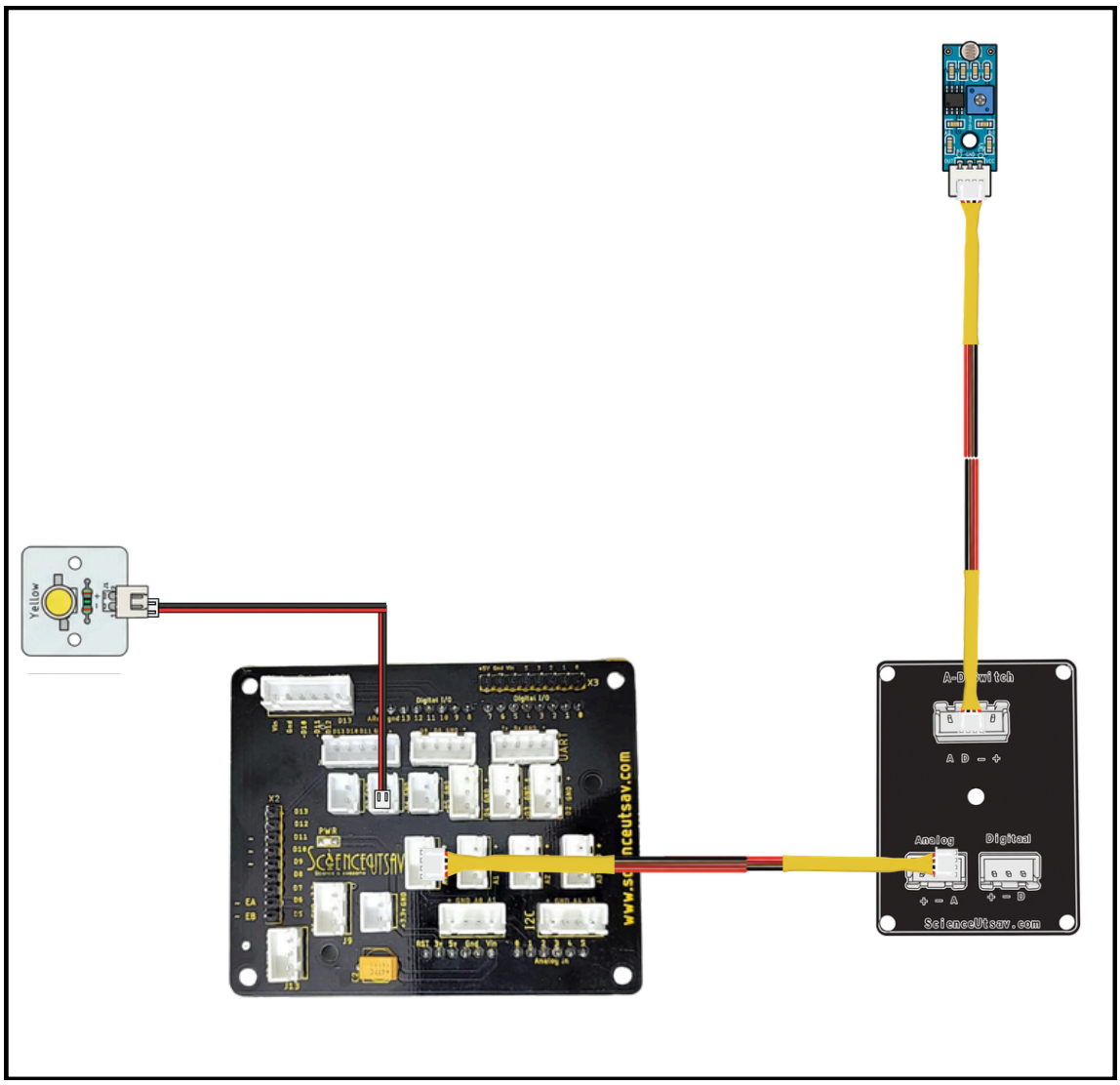
Step 8- Upload the code & close the super hero face with mask and observe the output

Code

```
// Session 1 Grade 7
// Iron Man / Lamp - LDR Auto 1W LED

const int ldrPin = A0;
const int led = 6;

int darkValue = 500; // adjust after testing

void setup() {
 pinMode(led, OUTPUT);
}

void loop() {
 int ldrValue = analogRead(ldrPin);

 if (ldrValue > darkValue) {
 digitalWrite(led, HIGH);
 } else {
 digitalWrite(led, LOW);
 }
}
```

INTERMEDIATE PROJECT

The circuit diagram of the project is given below

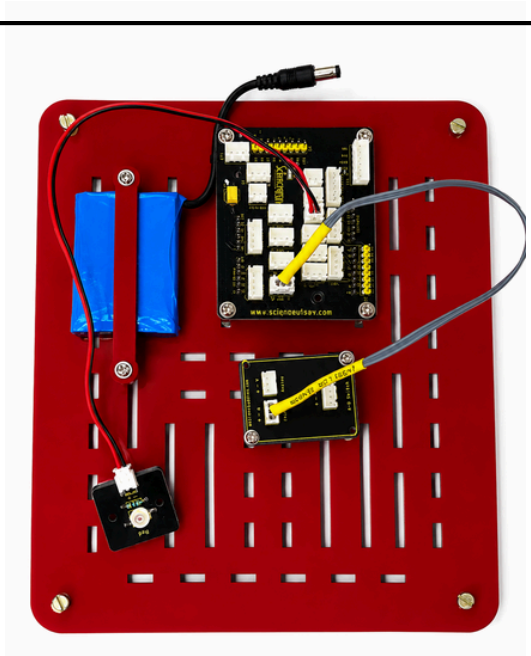


CONNECTION :

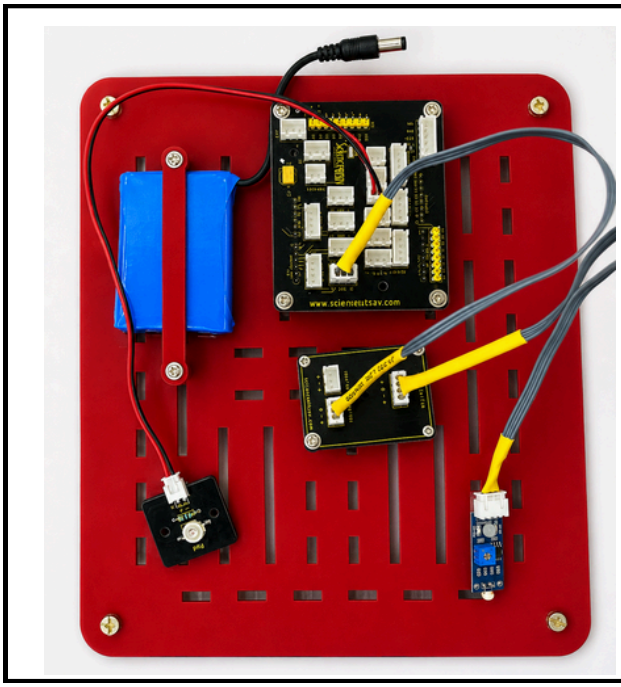
1.) A0 - AD SWITCH - LDR SENSOR

2.) D6 - 1W LED

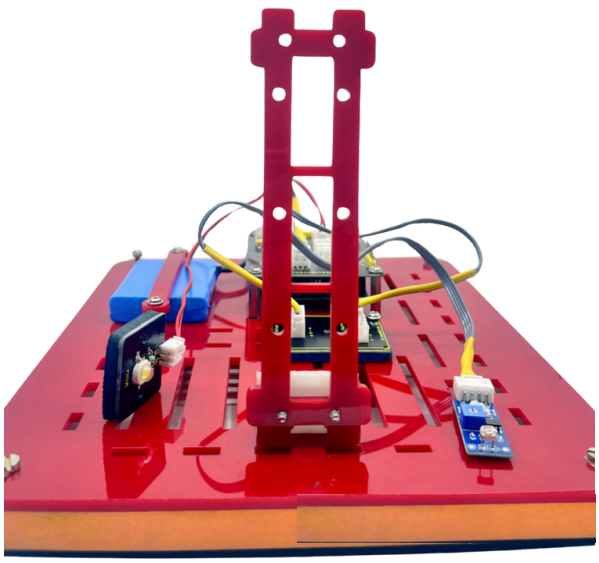
Step 1- Connect the AD switch and the LED to the shield



Step 2- Connect the LDR sensor to the AD switch

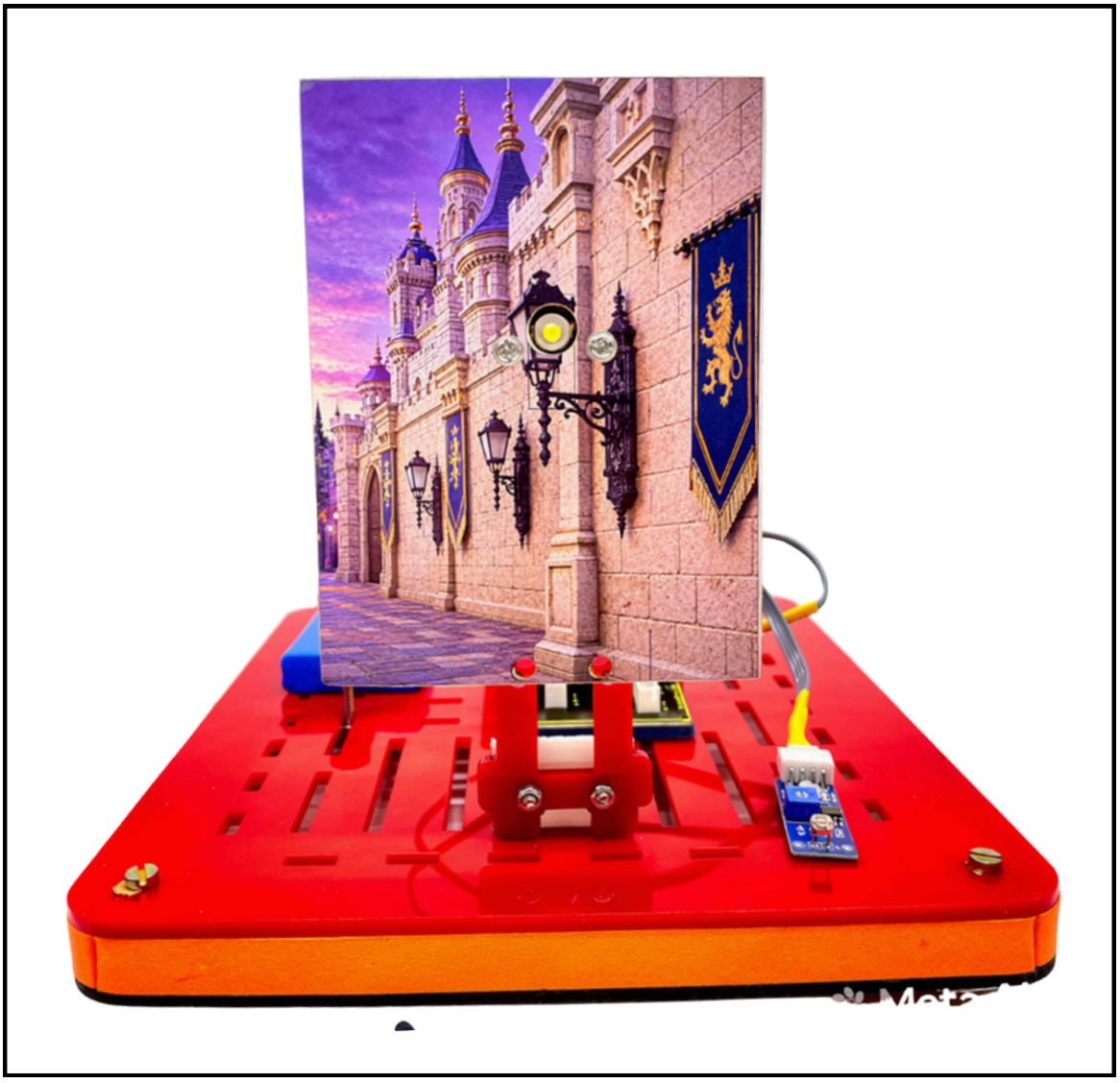


Step 3- Fix the structure as shown in figure



Step 4- Attach the template to the structure and fix the LED to it

Step 5- upload the code the code and observe the output

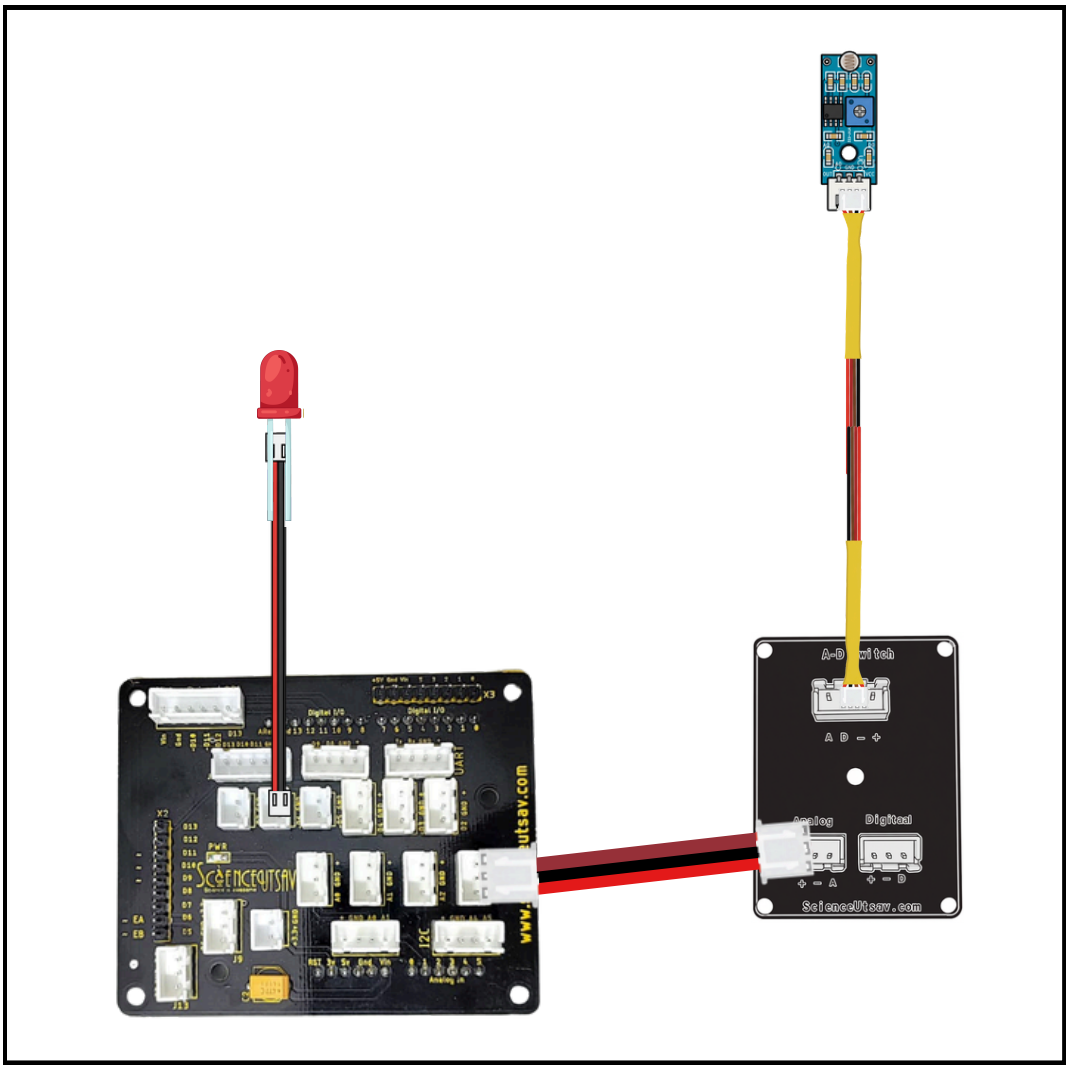


Code

```
// Session 1 Grade 8
// Lamp - LDR Auto Yellow LED

const int ldrPin = A0;
const int yellowLED = 6;

int darkValue = 500; // adjust after testing

void setup() {
 pinMode(yellowLED, OUTPUT);
}

void loop() {
 int ldrValue = analogRead(ldrPin);

 if (ldrValue > darkValue) {
 digitalWrite(yellowLED, HIGH);
 } else {
 digitalWrite(yellowLED, LOW);
 }
}
```

ADVANCE PROJECT

The circuit diagram of the project is given below

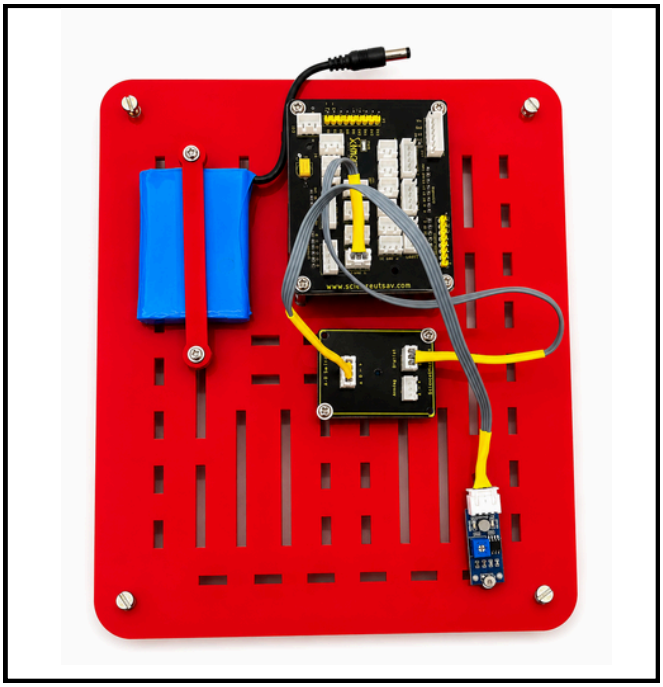


CONNECTION :

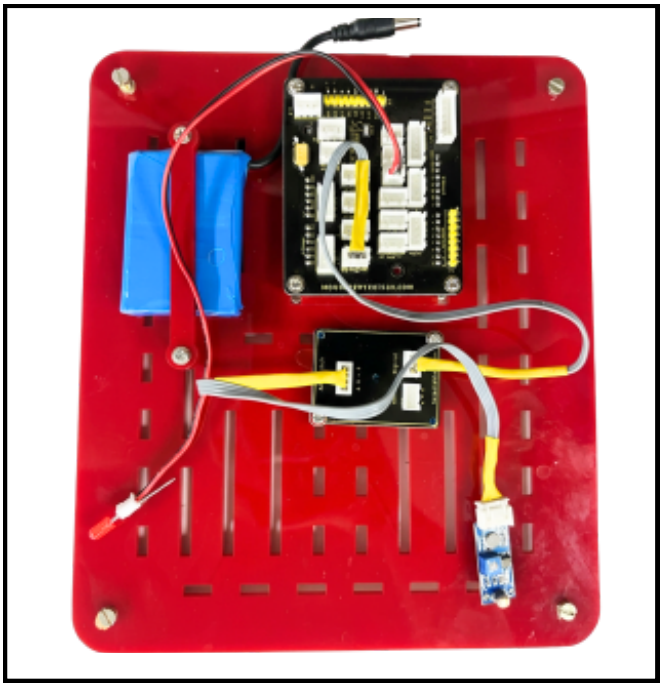
1.) A0 - AD SWITCH - LDR SENSOR

2.) D6 - LED

Step 1- Connect the AD switch and the sensor to it



Step 2- Attach the LED to the shield



Step 3- Attach the L-clamp to the template

Step 4- Attach the template to the the board

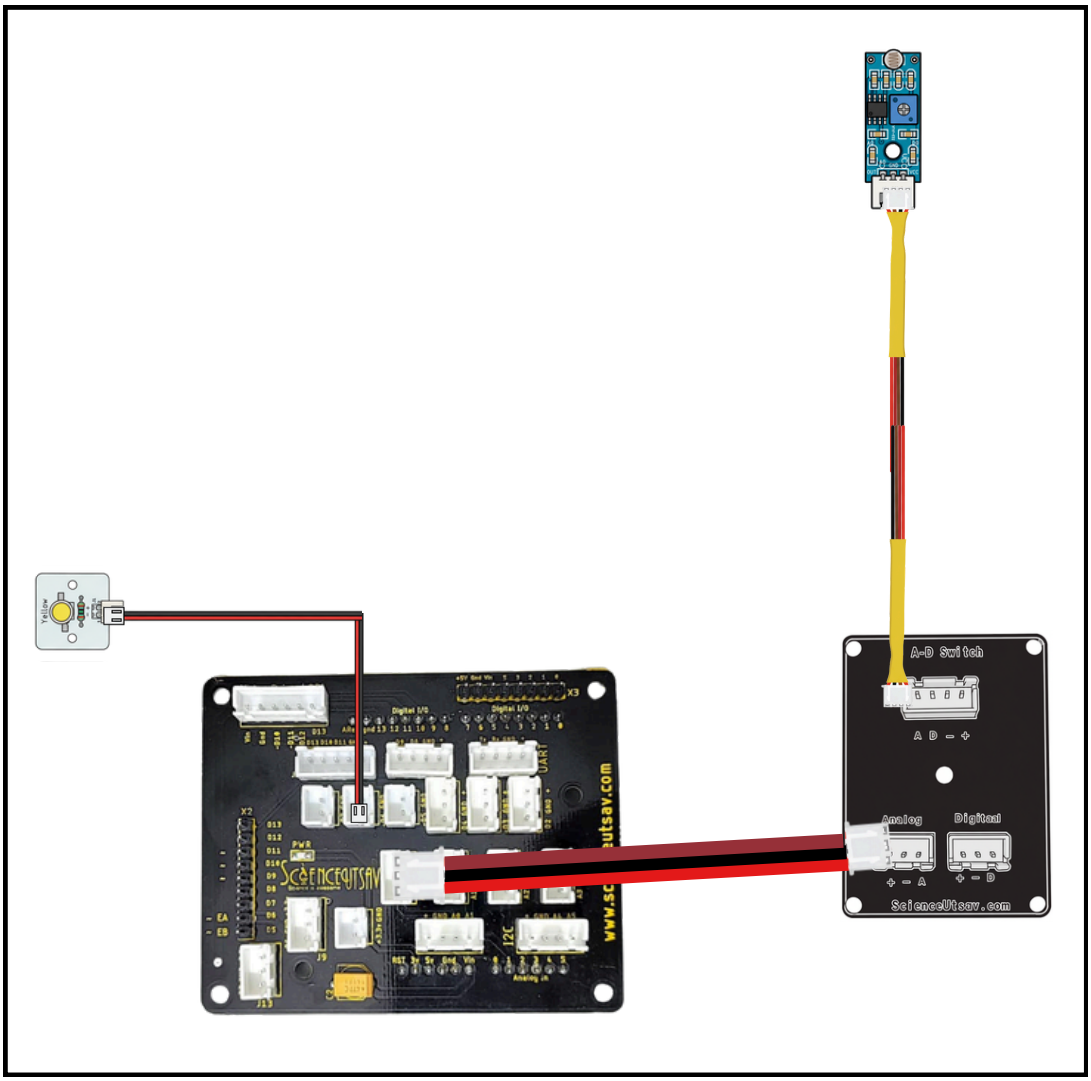
Step 5- Attach the LED to the template

Code

```
// Session 1 Grade 8
// Lamp - LDR Auto Yellow LED

const int ldrPin = A0;
const int yellowLED = 6;

int darkValue = 500; // adjust after testing

void setup() {
 pinMode(yellowLED, OUTPUT);
}

void loop() {
 int ldrValue = analogRead(ldrPin);

 if (ldrValue > darkValue) {
 digitalWrite(yellowLED, HIGH);
 } else {
 digitalWrite(yellowLED, LOW);
 }
}
```

ADVANCE PROJECT

The circuit diagram of the project is given below

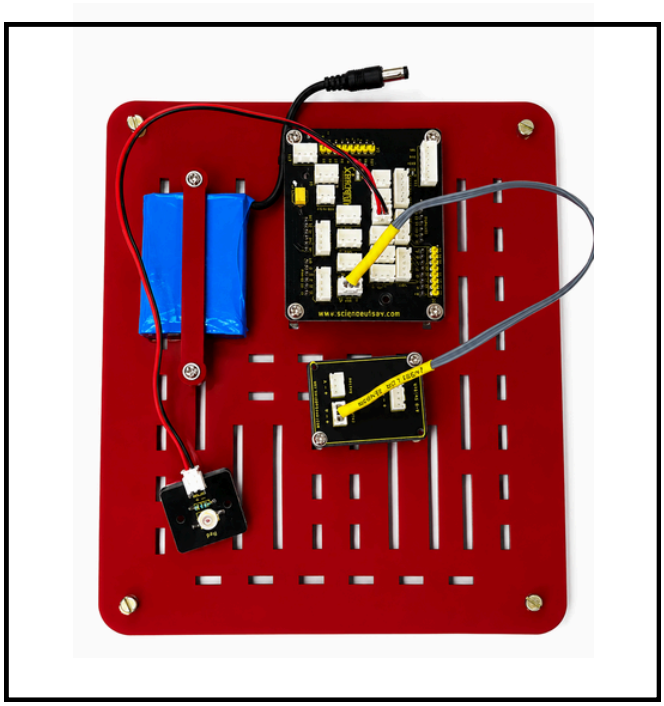


CONNECTION :

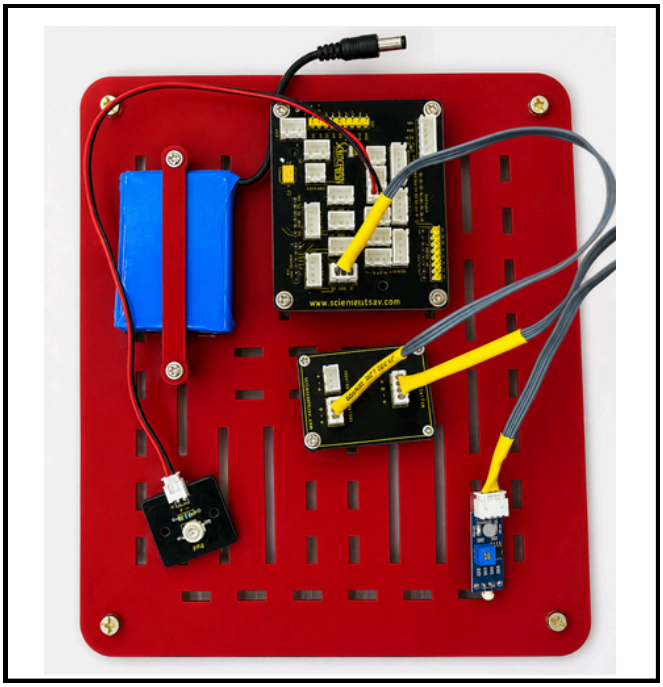
1.) A0 - AD SWITCH - LDR SENSOR

2.) D6 - 1W LED

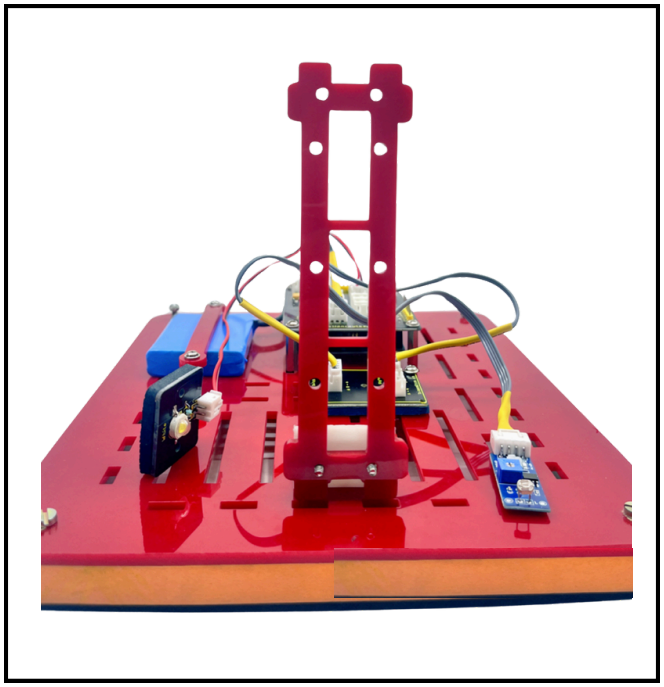
Step 1- Connect the AD switch and the LED to the shield



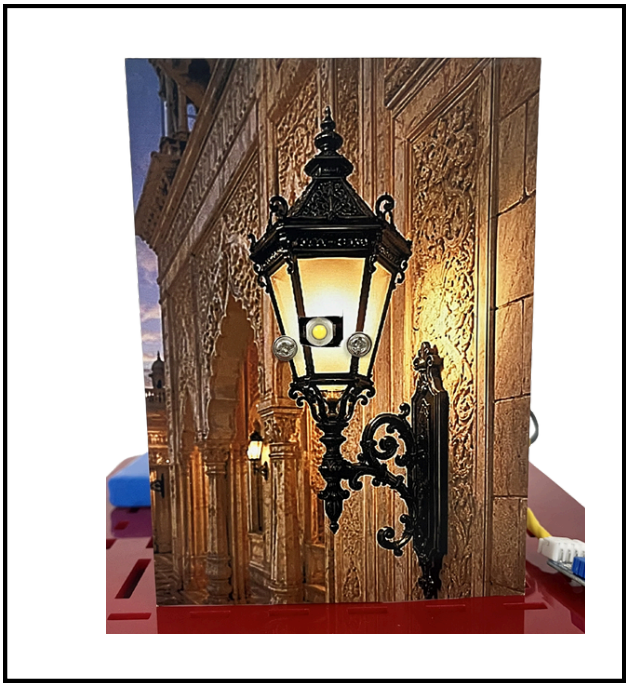
Step 2- Connect the LDR sensor to the AD switch



Step 3- Fix the structure as shown in figure



Step 4- Attach LED to the template



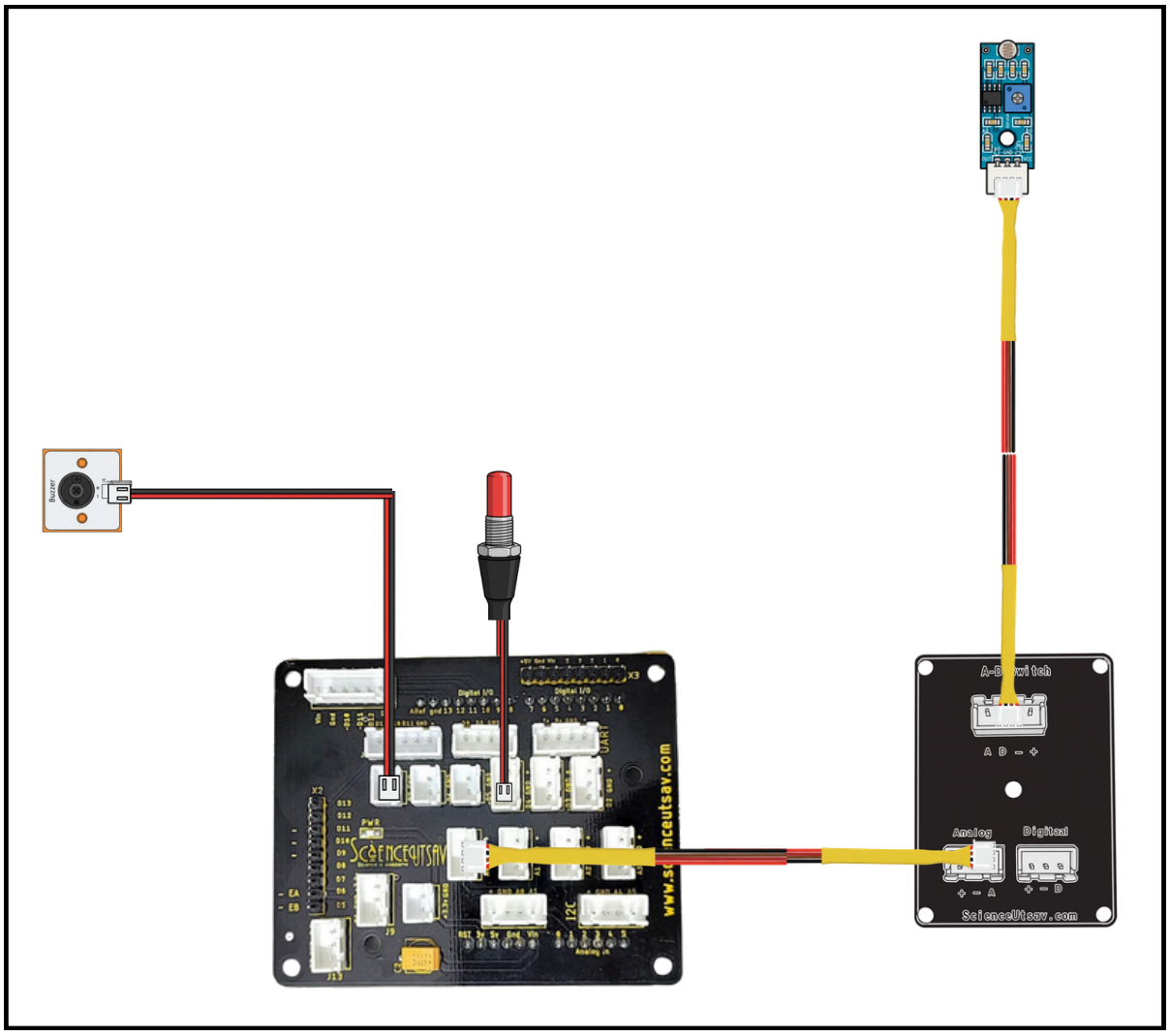
Upload the code and observe the output

Code

```
// Session 1 Grade 8
// Lamp - LDR Auto Yellow LED

const int ldrPin = A0;
const int yellowLED = 6;

int darkValue = 500; // adjust after testing

void setup() {
 pinMode(yellowLED, OUTPUT);
}

void loop() {
 int ldrValue = analogRead(ldrPin);

 if (ldrValue > darkValue) {
 digitalWrite(yellowLED, HIGH);
 } else {
 digitalWrite(yellowLED, LOW);
 }
}
```

EXPERT PROJECT

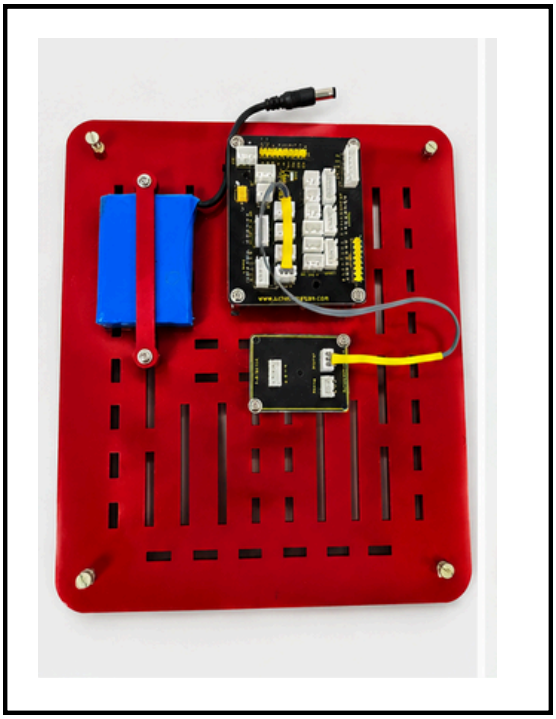
The circuit diagram of the project is given below



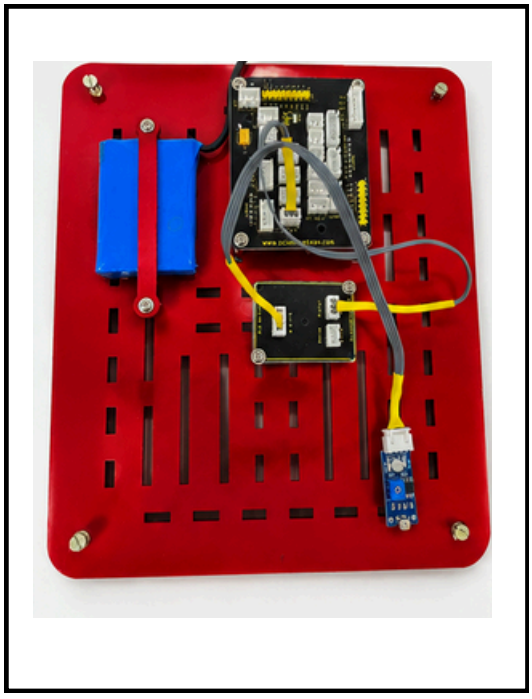
CONNECTION :

1. A0 - AD SWITCH - LDR SENSOR
2. D5 - PUSH BUTTON
3. D7- BUZZER

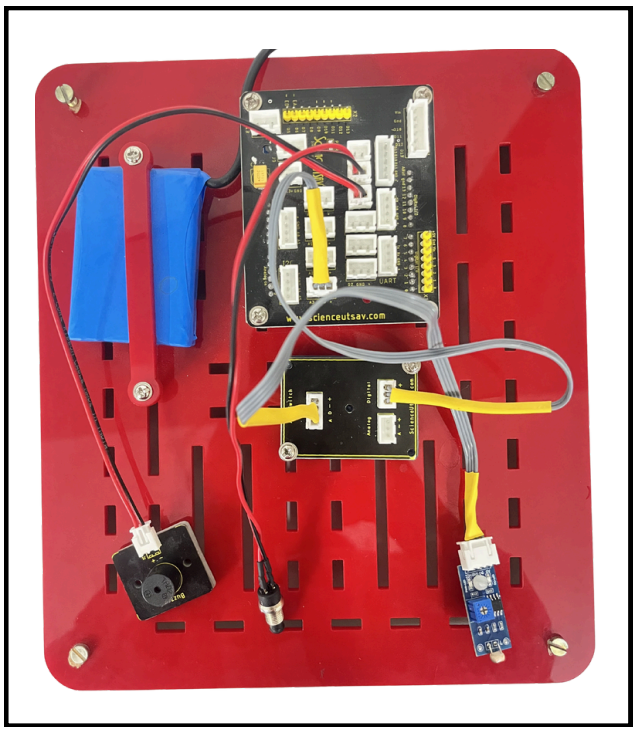
Step 1- Attach AD switch to the shield



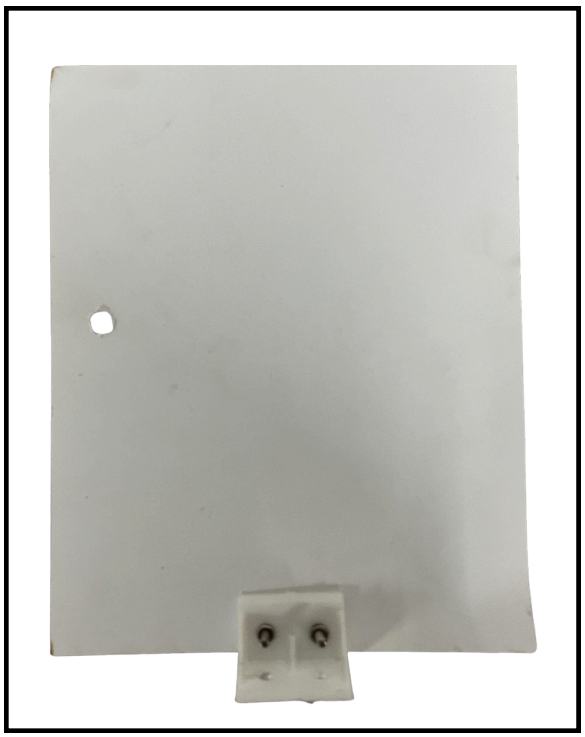
Step 2- Connect the sensor to AD switch



Step 3- Attach buzzer and push button as shown



Step 4- Attach L clamp to the template



Step 5- Fix the push button to the template

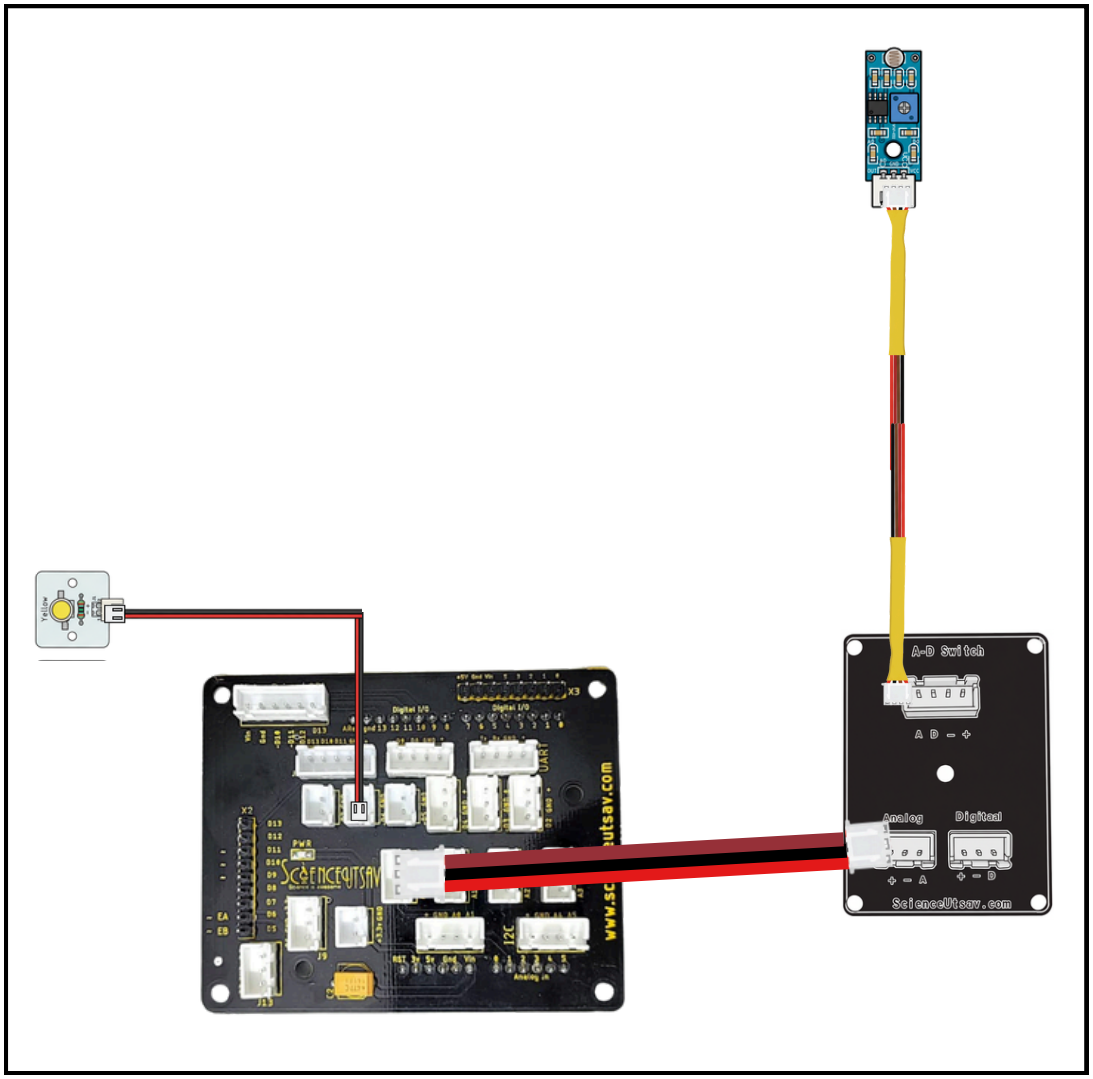
Upload the code and observe

Code

```
// Session 1 Grade 9
// Alarm - LDR + Buzzer + Push Button

const int ldrPin = A0;
const int pushButton = 5;
const int buzzer = 7;

int darkValue = 500; // adjust after testing
bool alarmStopped = false;

void setup() {
 pinMode(pushButton, INPUT_PULLUP);
 pinMode(buzzer, OUTPUT);
}

void loop() {
 int ldrValue = analogRead(ldrPin);

 if (digitalRead(pushButton) == LOW) {
 alarmStopped = true;
 digitalWrite(buzzer, LOW);
 delay(300);
 }

 if (ldrValue > darkValue && alarmStopped == false) {
 digitalWrite(buzzer, HIGH);
 } else {
 digitalWrite(buzzer, LOW);
 }

 // When light comes back, alarm can trigger again next time it becomes dark.
 if (ldrValue < darkValue) {
 alarmStopped = false;
 }
}
```

EXPERT PROJECT

The circuit diagram of the project is given below

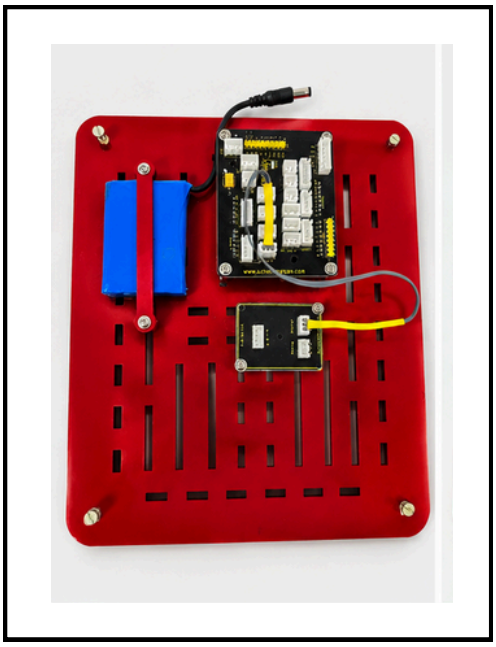


CONNECTION :

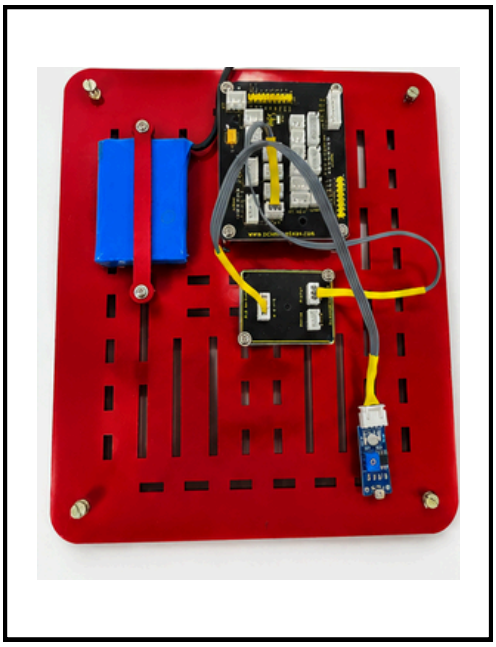
1.) A0 - AD SWITCH - LDR SENSOR

2.) D6 - 1W LED

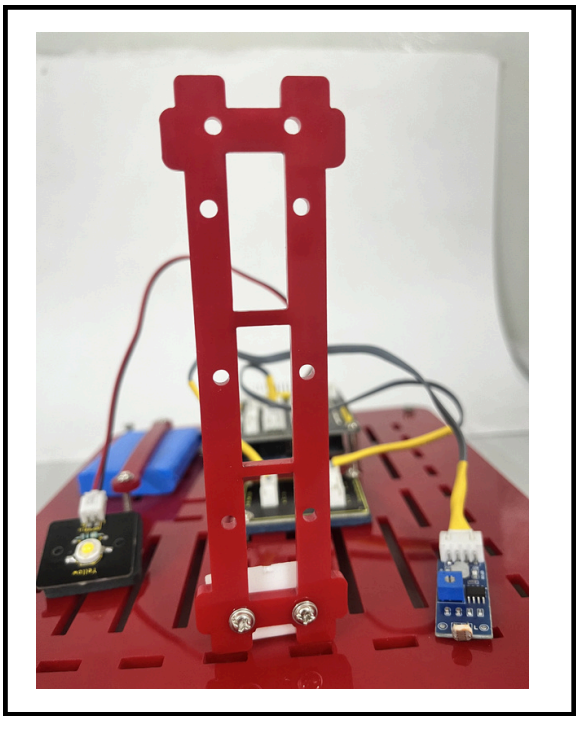
Step1- Connect the AD switch to the shield



Step 2- Connect the sensor to AD switch



Step 3- Fix the Structure as shown in the figure



Step 4- Attach the template to the stand and fix the LED to it

Upload the code and observe the output

Code

```
// Session 1 Grade 9
// Table Lamp - LDR + Yellow LED

const int ldrPin = A0;
const int yellowLED = 6;

int darkValue = 500; // adjust after testing

void setup() {
 pinMode(yellowLED, OUTPUT);
}

void loop() {
 int ldrValue = analogRead(ldrPin);

 if (ldrValue > darkValue) {
 digitalWrite(yellowLED, HIGH);
 } else {
 digitalWrite(yellowLED, LOW);
 }
```